

IN THIS ISSUE

News and Developments from Washington, D.C., Illinois, Massachusetts, New Mexico, Utah, France, India, Netherlands, Russia

COLD FUSION POLITICS..... 1,8
COLD FUSION UPDATES..... 2,3,4,6,8,10
COLD FUSION BUSINESS 3,6,9
COLD FUSION PROGRESS2,4

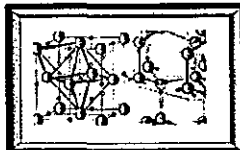
COLD FUSION EVENTS 1,8,10
COLD FUSION FUNDING 1
FOR THE WORLD 3,8
ICCF-5 CONFERENCE 6,9

MATERIALS CORNER 5,7
MORE INFORMATION..... 2,3,4,8
PEOPLE IN THE NEWS..... 3
TECHNOLOGY 1,2,4,5,7,9

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COLD FUSION TIMES

GOP LEADS U.S. INVESTIGATIONS INTO COLD FUSION

Washington D.C. - As the GOP leadership readies itself to slash the science budget by approximately one-third, there is hope at last for keeping cold fusion investigations alive and thriving. That hope lies in the House Science Committee. "I'm also especially interested in the so-called cold fusion area," said Rep. Robert Walker, new head of the House Science Committee (R., Pennsylvania) in the October, 1995 issue of *Popular Science*. He noted "there's something that's producing additional heat and it's being found in lab after lab. No one at this point can explain it. The hydrogen bill sets aside some monies to go into highly innovative research such as this". (Cont. Page 3)

FURTHER EVIDENCE FOR NUCLEAR PRODUCTS

New Mexico and India -- As discussed in the *Cold Fusion Times* for three years, helium is the major product of cold fusion reactions. Unlike hot (plasma) fusion, the production of tritium occurs in incredibly minute amounts, and in the absence of neutrons. However, if the electrodes are not pure and under other conditions, then trace amounts of tritium are produced. This issue presents the scientific evidence for nuclear events in CF with articles from Los Alamos and the Atomic Research Centre in India and other laboratories which document evidence for nuclear reactions in cold fusion. Dr. Claytor of Los Alamos describe the preparation of tritium using plasma loading into palladium at levels of nanocuries. Dr. Srinivasan at Bhabha Atomic Research Center (India) continues his series of papers from the previous issue of the *Cold Fusion Times* and discusses some of the protocols required to trigger nuclear events. (continued Page 5)

JAPAN FURTHER INCREASES INVESTMENT IN R&D!!

As Europe advances, and North America slowly awakens to the loss of this incredible energy technology, Japan has increased its ante into CF R&D with solid increases from circa \$30 million (see *COLD FUSION TIMES* v1n3) more than \$90 million.

EUROPEAN COLD FUSION PATENT ISSUES

Salt Lake City, Utah -- The European Patent Office has just issued a Notice of Intent for a patent to be granted on the original Pons-Fleischmann cold fusion patent. Unlike the U.S. patent system, the application now will undergo a nine month period during which those who claim prior art on the invention can register their protest. This notice of intent to grant the European patent is an important development because it shows again the seriousness of the field. It has taken quite a significant amount of time for this development but it heralds a reversal of the pendulum of "conventional wisdom".

COLD FUSION MERGES FUSION COMMUNITIES

Champaign, Illinois - Cold fusion contributed to the 16th biannual Symposium on Fusion Engineering (SOFE Oct. 1995) sponsored by the IEEE (Institute of Electrical and Electronic Engineers), the American Nuclear Society, and the University of Illinois. Although most of the conference was devoted to "hot" or plasma fusion topics like magnetic field and blanket technologies, plasma physics, fusion neutronics, tritium breeding and ITER, the field of cold fusion was represented. There were three papers on cold fusion, including a cold fusion demonstration. (continued Page 2)

Try to identify the photograph on the right. It is an image of a coiled object and has great importance.

Hint: It is not a Garter Snake. Nor is it an "Out-break" virus. For answer see page 7 for details.



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TOWARD THE FUTURE

CHEMICALLY ASSISTED NUCLEAR REACTIONS

by Edmund Storms

"More than six years have passed since the modern era of 'cold fusion' was started by Profs. Stanley Pons and Martin Fleischmann (then at the University of Utah). Their claims of being able to produce nonpolluting energy from a renewable source using a simple apparatus created great initial excitement. However, difficulties in repeating the work, combined with the absence of any acceptable explanation, caused most scientists to conclude that the claims were based on delusion. Nevertheless, some people continued to explore the possibilities.

Criticisms made by skeptics were taken seriously, errors have been reduced or eliminated, and a wide variety of studies have been done using very modern equipment in many countries. The early problem of reproducing the effect has been largely eliminated, nuclear byproducts have been found, and theoretical explanations abound. The problem now is more psychological than scientific. In spite of this new and improved information, general skepticism about the effect continues within the scientific community and general rejection by the U.S. and many other governments remains unchanged.

Nine international conferences have been held and several professional societies have included sessions about cold fusion, the most recent being the American Chemical Society. The literature on the subject has grown to over 1300 publications, many peer reviewed

by major scientific journals. ... The field has expanded from claims of d-d fusion being produced in palladium using electrolysis to at least ten different method-environment combinations. These environments include normal hydrogen as well as deuterium . . . Sufficient energy has been observed to encourage commercial development. The present status of the field will be summarized with respect to what has been discovered; where work is being done; and how this new field is expected to affect conventional thinking."

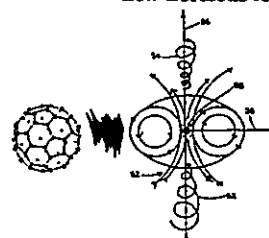
[This abstract by Dr. Storms, presented at the IEEE/NPSS SOFE '95 meeting, summarizes the field. It also is a vectored direction for what is becoming the conventional thinking of those scientists who actually examine the literature in cold fusion.]

COLD FUSION MERGES FUSION COMMUNITIES

Continued from page 1

Papers were given by Prof. Peter Hagelstein (MIT), Dr. Kohji Kamada (National Institute for Fusion Science, Nagoya), and Dr. Edmund Storms (Los Alamos- retired). Dr. Storms also contributed to the panel discussion. The demonstration, done by a group from the University of Illinois, was a CETI light water device similar to the one successfully demonstrated at the Fifth International Conf. on Cold Fusion (ICCF5). (continued on page 5)

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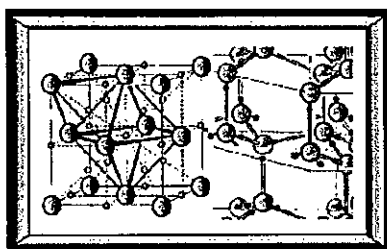
Rep. Walker - Popular Science

Cold Fusion Times LOGO - Palladium must be fully filled
with deuterons to generate the reactions. The logo was in part
based upon a water structure drawn by John Mara. In the logo,
the face-centered cubic fully loaded metallic structure is shown
on the left, totally filled with deuterons obtained from the heavy
water on the right hand side.

Image Processing -- JET Technology

COLD FUSION RADAR

returns next issue -
stay tuned for the latest in
information available in this field



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The Journal Of The Scientific Aspects Of
Loading Isotopic Fuels Into Materials

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PEOPLE IN THE NEWS Fall '95

GOP LEADS

(Continued from Page 1)

Representative Walker discussed the anticipated 35% budget cut imminent for hot fusion research this year. He also supports CF research even if it is not classic fusion in the true sense of the words. In the new and improved view of Congress, "hydrogen" represents THE alternative energy source.

Representative Walker is scientifically correct to examine what CF scientists have known for six years. He should be, and will be, praised for re-vectoring the United States energy potential at this critical time.



Rep. Robert Walker (R, Pa.)
Chairman House Science Committee:
"Hydrogen seems to be a potential energy resource with benign environmental properties."

THE CURRENT SHIFT IN ENERGY GENERATION

Mitchell Swartz

The times they are indeed a'-changing. We now know that fission has distinct disadvantages. Public opinion, mishandled disasters, and growing spent fuel rods remain expensive obstacles and problems. With 109 nuclear power plants supplying a fifth of the energy of the United States, there has been no new plants built since 1989. Fission research is dying -- heralded by the fact that the DOE only spent \$40 million this year for reactor technology. "The countries where nuclear power has succeeded are technophilic and resource poor, like France, Japan, and South Korea", as was discussed in the *Economist* (9/16/95) recently. So where to go?

Hot fusion is incredibly tardive, if not doomed, due to its inordinate gestation time, hypercomplexity, and requisite temperature. Even physicists and engineers in the hot fusion field agree that the technologies required for hot fusion will not be developed effectively to be useful until at least well into the next century. Worse, the neutrons produced by hot fusion (and not produced by cold fusion) cripple the very walls of fusion reactors, if breakeven is ever achieved.

Although small by volume, the barriers create considerable handling problems. The walls comprising ITER (International Thermonuclear Experiment Reactor resulting from a Soviet-American Geneva summit in 1985) are planned with a mere three year (or less) replacement time. Each "molt"/replacement will produce yet another hundred tons of nuclear waste. So serious are these problems, that the planned forecast for ITER has been delayed by another three years, coupled with the requirement of another \$4 billion. Furthermore, it will lead to hot fusion in the time range, now down-sized, of only 50 to 100 seconds! Cold Fusion has already exceeded that time for breakeven!!

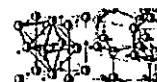
Congress meanwhile is growing exhausted from the soaring funding and dwindling returns. When interviewed by *Physics Today* (9/1995), one senate staffer quipped, "Those (hot) fusion fellas remind me of Indiana Jones. Whenever they face a perilous (funding) situation, they come up with a miraculous solution". In contrast, the cold fusioners pursue their research and development quietly with a paucity of funding. Those who continue to study and work in this field know that cold - or solid state - fusion remains the hope of the world for the safe generation of energy by aneutronic processes.

OF GREAT INTEREST
TO OUR READERS
SEE FOR THE WORLD (PAGE 8)

COLD FUSION TIMES

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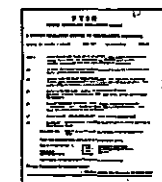
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Information Update for "Newbies" (Newcomers)

There are always incoming requests for more information about cold fusion. Supplementing our own references, previously listed here, are some picks from Mr. Jed Rothwell. He has generously offered to supply e-mail copies of several of these papers via the Internet (see box below for more information).

The availability of the other publications can be found at the telephone numbers and locations listed below. We both thank and applaud Jed Rothwell, winner of our 1995 ICCF-5 Award (previous issue), whose interesting and informative comments are read worldwide.

Jed Rothwell's RECOMMENDED REFERENCES ON COLD FUSION

"In March, 1989 a new source of energy popularly known as 'cold fusion' was announced by Professors Martin Fleischmann and Stanley Pons. This caused a great media brouhaha, but there have been few articles about the subject in U.S. media since then. A great deal of work in this field has been done. While progress is frustratingly slow in some ways, the technology still looks extremely promising.

Most of the R&D is now being done in Japan by Japanese corporations, which tend to keep their results under wraps. For example, we have heard little from Pons and Fleischmann over the last three years because they are now in charge of the Toyota IMRA research project in Nice, France, and they are official advisors to the Ministry of International Trade and Industry (MITI) R&D project. Toyota and MITI like to play their cards close to the vest."

"I would like to encourage anyone who is interested in this subject to read the scientific literature. A great many papers have been published in the last 6 years. There are so many papers that before jumping into the subject, I recommend you read some broad reviews and books, particularly the ones written by Mallove, and Storms. Here are some recommended publications to get started with. Items marked [E-Mail] are available from me e-mail or diskette. Items marked [SCIENCE Lib 2] can be downloaded from the CompuServe SCIENCE forum physics library 2." —

Jed Rothwell can be reached at
Cold Fusion Research Advocates
2060 Peachtree Industrial Court, Suite 313 Chamblee,
Georgia 30341 Tel: 770-451-9890 Fax: 770-458-2404
E-Mail: CompuServe 72240,1256
or JEDROTHWELL@DELPHI.COM]

GENERAL

Fire from Ice: Searching for the Truth Behind the Cold Fusion Furor, (John Wiley & Sons, May, 1991), by Dr. Eugene F. Mallove. The definitive book on the subject.

E. Storms, "Cold Fusion Heats Up," MIT Technology Review, 20-29, May-June 1994

E. Storms, "A Critical Review of the 'Cold Fusion' Effect," submitted to Physical Review B (1995) 67 [Preprints available from me]

Infinite Energy Magazine, Edited by E. Mallove, P.O. Box 2816, Concord, NH 03302-2816, Tel: 603-228-4516, Fax: 603-224-5975 E-mail: 76570,2270

Information on cold fusion can be found in the John Logajan's World Wide Web home page: URL = <http://www.skypoint.com/members/jlogajan>

J. Rothwell, "Highlights of the Fifth International Conference on Cold Fusion," 14 pages, [E-Mail] [SCIENCE Lib 2]

TECHNICAL

Cold Fusion Times, by Mitchell Swartz, P.O. Box 81135, Wellesley Hills, MA 02181 E-mail address: mica@world.std.com

Fusion Technology, a technical journal published by the ANS has published many articles about cold fusion. Contact: Publications Manager, The American Nuclear Society, 555 North Kensington Ave, LaGrange Park, IL 60525. Back issues of Fusion Technology are available from the APS publications office at 708-352-6611.

Fusion Facts, a monthly newsletter. Contact subscription office at: P.O. Box 48639, Salt Lake City, UT 84158. Tel: 801-583-6232 Fax: 801-583-6245. Fusion Facts publishes the diskette bibliography I referred to above.

The Fifth International Conference on Cold Fusion (ICCF5) was sponsored by IMRA and held in Monte-Carlo, Monaco April 9 - 13, 1995. The proceedings will be published by IMRA Europe, S.A., Center Scientifique, B.P. 213, 22, rue Albert Caquot, 06904 Sophia Antipolis Cedex, France. Mallove offers a set 4 videotapes "Highlights of ICCF5." This is over 8 hours long and it includes all abstracts and an index of participants. Contact: 76570,2270@compuserve.com.

The Proceedings of the Fourth International Conference on Cold Fusion (ICCF4). This conference was sponsored by the Electric Power Research Institute (EPRI), Advanced Nuclear Systems, and by the U.S. Office of Naval Research. It was held December 6 - 9, 1993, at Hyatt Regency Maui, Lahaina, HI. The proceedings can be purchased from: EPRI Distribution Center * 207 Coggins Drive * P.O. Box 23205 * Pleasant Hill, CA 94523 * Tel: 510-934-4212

Another version of the ICCF4 proceedings was published by the American Nuclear Society: Proceedings of the Fourth International Conference on Cold Fusion, Dec. 6 - 9, 1993, Transactions of Fusion Technology, 1993, Vol. 26, No. 4T, Part 2 (Dec. 1994), ISSN: 0748-1896. This is a peer-reviewed set of some of the papers.

Frontiers of Cold Fusion, ed. H. Ikegami. The proceedings of the Third International Conference on Cold Fusion (Nagoya, Japan, October 21 - 25, 1992) in Nagoya, Japan. Available from Universal Academy Press, Inc., PR Hogo 5 Bldg., 6-16-2, Hongo, Bunkyo Tokyo 113, JAPAN. Tel. 011-81-3-3813-7232, Fax: 011-81-3-3813-5932. Price 22,000 yen (U.S. \$194.77, Air shipping: \$26.65)

P. Hagelstein (M.I.T.), "Summary Of Third International Conference On Cold Fusion In Nagoya," 43 pages, \$5 [E-Mail] [SCIENCE Lib 2]

The Science of Cold Fusion, ed. T. Bressani. The proceedings of the Second Annual Conference On Cold Fusion. (Como, Italy, June 29 - July 4, 1991); contact: SIF, Via L. degli Ondalo 2, 40124 Bologna, ITALY. From the Second Annual Conference proceedings, we recommend: M. McKubre (SRI), "Isothermal Flow Calorimetric Investigations Of The D/Pd System," p. 419 - 443

M. McKubre et al., "Isothermal flow calorimetric investigations of the D/Pd and H/Pd systems," J. Electroanal. Chem. 368 (1994) 55

M. H. Miles (Naval Air Weapons Center), B. F. Bush (SRI), D. E. Stillwell (CAES), "Calorimetric Principles and Problems in Measurements of Excess Power during Pd-D2O Electrolysis," J. Phys. Chem. 1994, 98, p. 1948-1952

M. Fleischmann (Univ. Southampton), S. Pons (IMRA Europe), "Calorimetry of the Pd-D2O system: from simplicity via complications to simplicity," Physics Letters A, 176 (1993) 118-129
E. Storms (Los Alamos), "Review of Experimental Observations About The Cold Fusion Effect," Fusion Technology, Vol. 20, Dec. 1991 433 - 477. Dated, but still a superb technical introduction to the field.

O. Reifenschweiler (Phillips), "Reduced radioactivity of tritium in small titanium particles," Physics Letters A, 184 (1994) 149-153

M. H. Miles and R. A. Hollins (Naval Air Weapons Center), B.F. Bush and J.J. Lagowski (Univ. Texas), "Correlation of excess power and helium production during D2O and H2O electrolysis using palladium cathodes," J. of Electroanalytical Chemistry, 346 (1993) 99 - 117.

H. Gerischer (Fritz Harber Institute Der Max Planck), "Memorandum On The Present State Of Knowledge On Cold Fusion." [E-Mail] [SCIENCE Lib 2]

MEDIA COVERAGE

BBC "Horizon" series science documentary, "Too Close to the Sun." Broadcast in the U.K. and Canada, March 1994.

Popular Science, August 1993 issue, "COLD FUSION Fact or Fantasy," by Jerry Bishop, cover story

Sunday Times (U.K.), June 27, 1993, "Nuclear confusion," by Neville Hodgkinson

The National Public Radio (NPR) program "Science Friday" on June 25, 1993 was devoted to cold fusion. It was moderated by Ira Flatow. Panelists included Michael McKubre of SRI, John Huizenga of Rochester University, Peter Hagelstein of MIT, Melvin Miles of the Naval Air Warfare Center, and Bruce Lewenstein of Cornell University. For a tape, send \$12.50 to: NPR Tapes * Washington, DC 20036 * Visa orders: 202-822-2323. Specify the date (06/25/93)

The NPR program "Science Friday" devoted to cold fusion January 20, 1995.

The Canadian Broadcasting Company (CBC) superb documentary on cold fusion on June 24, 1993, "The Secret Life of Cold Fusion."

New York Times, November 17, 1992, "Cold Fusion, Denied in U.S., Is Hot in Japan," by Andrew Pollack, p. B5

ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

MATERIALS CORNER

This issue: TRITIUM PRODUCTION FROM COLD FUSION

Some Experiments on the Variation of the Radioactivity of Tritium Absorbed by Titanium

Otto Reifenschweiler, Philips Research Laboratories

Eindhoven, The Netherlands -- Small amounts of tritium are sorbed by titanium preparations consisting of small monocrystalline particles (ϕ approx 15 nm) obtained by gas evaporation. In a first experiment, a TiT(0.0035) preparation is heated and the radioactivity, A , is measured via the X-radiation detected by a GM-tube. The radioactivity decreased from the initial value down to 72% at 160 deg. C, further decreased to 60% at 275 deg. C and increased again to the initial value at 360 deg. C before decomposition of the preparation.

It is shown that, during the whole temperature trajectory up to 360 deg. C, no loss of tritium has occurred and that the

decrease of the count rate is entirely due to the decrease of tritium radioactivity. A detailed analysis of the generation of the X-radiation shows that it is not possible to explain the course of $A=f(t)$ by changes from chemi-absorption at the surface to bulk absorption of the tritium in the small Ti particles as has been claimed recently.

In a second important heating experiment with a TiT(0.035) preparation, the tritium is desorbed during heating on account of the 10 times higher concentration and the tritium released from the preparation is measured by a second GM-tube. By addition of the two radioactivities, that remaining in the solid and that expelled as a gas, it is shown that the course of $A=f(t)$ is equivalent for both experiments carried out under very different conditions.

In a further heating experiment with a TiT(0.065) preparation where the surface of the Ti-particles is oxydized, a course of $A=f(t)$ of the same kind as that for the first experiment is obtained. The evaluation of these three heating experiments done under very different conditions gives a high degree of evidence for the decrease of tritium radioactivity. There is strong suspicion that the two different effects, cold DD fusion and the decrease of tritium radioactivity, are caused by the same or two related fundamental principles.

Tritium Production from a Low Voltage Deuterium Discharge on Palladium and Other Metals

T.N. Claytor, K.G. Tuggle and D.D. Jackson
Los Alamos National Laboratory

Los Alamos, NM - Over the past year we have been able to demonstrate that a plasma loading method produces an exciting and unexpected amount of tritium. In contrast to electrochemical hydrogen or deuterium loading of palladium, this method yields a reproducible tritium generation rate when various electrical and physical conditions (gas pressure, gas purity, initial palladium metallurgy, etc.) are met. Small diameter wires (100-250 Microns) have been used with typical gas pressures above 200 torr at voltages and currents of about 2,000V at 3-5 A. By carefully controlling the sputtering rate of the wire, runs have been extended to hundreds of hours allowing a significant amount ($> 10^5$ nCi) of tritium to accumulate. No pre- or post-gamma activity or neutron emission (to 1 C/hr) has been detected in any of the samples. We will show tritium generation rates for deuterium-palladium foreground runs that are up to 25 times larger than hydrogen-palladium control

experiments using materials from the same batch. The reproducibility has allowed us to vary parameters that have been difficult to investigate with previous methods. We intend to illustrate the difference between batches of annealed palladium and as received palladium from several batches to demonstrate that the tritium generation rate can vary by a factor of 40 from batch to batch. The effect of other metals (Pt, Ni, Nb), gas pressure, wire and plate thickness on the tritium generation rate will be shown. We plan to discuss these new procedures (including the effect of gas impurities, and palladium purity), present typical results, and speculate concerning the implications for further work.

COLD FUSION MERGES FUSION COMMUNITIES

(Continued from page 2)

As such it was described briefly in the last issue of **Cold Fusion Times**. The CETI cell has been reported to have been run for seven weeks, producing a steady 20 watts excess, for an 85 megajoule excess heat. This is an order of magnitude more energy than the biggest hot fusion test on record produced, at the Princeton Plasma Physics Laboratory (PPPL) in 1994.

The **COLD FUSION TIMES** needs your subscription, and support. It is the only cold fusion journal which is devoted exclusively to the material, economic, and intellectual property issues of loading isotopic fuel - a hydrogen isotope - into a material to produce useful energy.

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THE PATENT PAGE

Analysis of the Intellectual Property
back next issue

COLD FUSION THEORIES

[This issue continues with a few excerpts from the Highlights of the Fifth International Conference on Cold Fusion (ICCF-5) by Jed Rothwell. The was the **COLD FUSION TIMES WINNER for the BEST ICCF-5 REPORT.** Mr. Rothwell has served the scientific community as he has translated and reported on world-wide events in the field of cold fusion. His thoughtfulness underlines the pragmatic issues confronting this field. Readers are referred to his full articles posted on the Internet (sci.physics.fusion) and in *Infinite Energy*.]

THE BUSINESS OF COLD FUSION

by Jed Rothwell

Monaco - On the last day, Klein [2], of Bechtel Corporation, gave a superb talk on the economics and ABCs of developing cold fusion into a practical form of energy. He pointed out that solar photovoltaic cells use zero cost energy, but they still cannot compete with conventional sources because the fuel cost is not the only economic factor. He said that a cold fusion power reactor might be economical at the large scale, like today's coal or fission; or it might be economical at the substation level; or it might even be economical in home generator units operating at 10 or 20 kilowatts, in which case people will gradually unplug from the power distribution network.

This is bold speculation, and Klein is to be commended for talking about it, because his corporation's main line of work is in large scale energy installations like oil refineries and power stations, so he is speculating about something that may put him out of business. He did not describe a logical conclusion to this train of thought. There is no reason to suppose that miniaturization will stop at the level of a 20-kilowatt home reactor.

"Washing machines, coffee pots, children's toys, earphones and pacemakers may someday have built-in CF power."

Klein discussed many technical requirements for a practical source of energy, such as the need for a method of throttling the energy. It would be nice if we could turn on the flow of energy, turn it up, down, and then quench it. I would like to add a comment: while this is generally true, there are conventional sources of energy that cannot be quenched, and that can only be throttled to a certain extent. The best example is a ton of burning coal: once you light it, you have to let it burn. That is why coal is relatively inflexible and limited to large scale applications where continuous energy is needed and instantaneous, fine control over the reaction rate is not required. A scientist might think that the issues Klein discussed are premature, but they are not. If everyone in this field would move these issues to the top of his agenda, progress would be swift, and the field would be inundated with funding.

Over the years many excellent alternatives to pure palladium have emerged: thin film [3], palladium black [10], light water [3, 16, 17], ultrasound [7, 8, 9], proton conductors [18]. Other methods, like sparking [19] and glow discharge [20, 26], have not been as widely replicated, but they show promise. Yet the

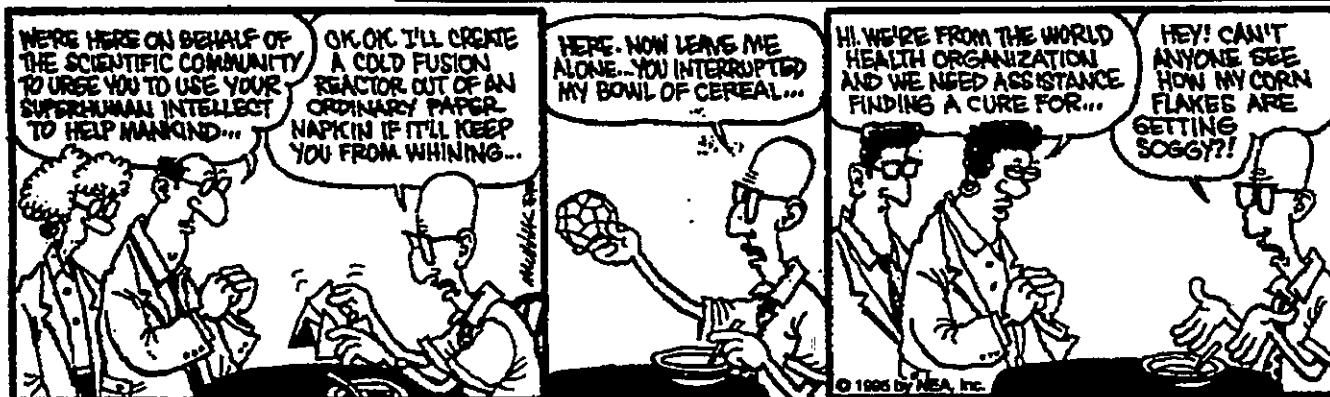
majority of scientists in the field ignore these promising approaches and continue using only palladium. Instead of selecting the easiest and most successful methods, they insist on using the oldest, least effective, and most frustrating technology, as if they were computer scientists who insisted on building a vacuum tube machine in the age of transistors. Many papers describing the heartbreaking difficulties these people face: the high loading, surface treatments, problems with cleanliness, and the many heroic techniques they are forced to employ. Kunitatsu [21] and others continue to search for ways to improve loading in palladium with electrolysis, instead of using other methods in which loading does not matter. Scientists who use pure palladium must wait 30 days or longer for a tiny, marginal, excess heat reaction to flicker on -- a reaction which often abruptly dies. Contrast this with the E-Quest device, which turns on in a fraction of a second and produces a 300% excess, or the Patterson CETI cells, which turn on in 20 minutes and produce up to 1000% excess. Okamoto [22] reported that the NEDO Icarus program, for example, saw only two excess heat reactions during the entire year, peaking at 16% excess. Six years of low level results have failed to convince mainstream scientists that CF is real. I asked Okamoto about the Icarus results, and he said their calorimetry is excellent, so they have no doubt that the 16% excess is real. (Okamoto is at the Tokyo Inst. of Technology, not in the Icarus project.)

As I expected, Pons and Fleischmann [23] did not reveal any details about their recent work. They have not revealed much since 1992, even though they have achieved some spectacular successes since then, including long boiling events.

(continued Page 9)

ROBOTMAN by Jim Meddick

COLD FUSION TIMES' FAVORITE CARTOON OF THE "MONTH" Boston Herald (8/16/95)



Observation of High Multiplicity Bursts of Neutrons During Electrolysis of Heavy Water with Palladium Cathode

A. Shyam, M.Srinivasan, T.Kawshik and L.Kulkarni, Bhabha Atomic Research Centre

Trombay, Bombay, India-

Experiments were conducted to detect neutron bursts during the electrolysis of D_2O (LiOD) and H_2O (LiOH) using tubular palladium cathodes (area - 300 cm^2) and nickel anode cell (Milton Roy) at low current densities (20 to 65 ma/cm^2). Data was acquired for 2 months, of which 15 days each were devoted to "control" experiments before and after the D_2O run.

The annular neutron detection system comprised of a bank of 16 BF_3 tubes embedded in a polyethylene cylindrical thermalising assembly in which an instantaneous neutron burst is temporally stretched to a few tens of micro-sec duration. The electrolytic cell was placed at the centre of this assembly. The preamplifiers and detector connections were specially designed so as to be electromagnetically and hermetically sealed. It was confirmed that the setup did not give spurious counts even under the high humidity conditions of the Bombay monsoon season. The overall neutron detection efficiency of the system was approx. 10% with a background of approx. 0.049 ± 0.002 counts/sec over a 15 day period.

The direct and dead-time (100 micro-sec) filtered train of pulses was continuously counted in 5 sec intervals and the data recorded on a personal computer. The total neutron counts per day with the D_2O cell were found to be consistently about 9% above the background. A significant observation emerged from the frequency distribution of 5 sec counts which was close to Poisson in case of background but contained several large multiplicity events in presence of the H_2O and D_2O cells (Fig. 1) implying the emission of bursts of 20 to 100 neutrons in < 100 micro-sec duration from the latter. The feature is more evident from the multiplicity distribution of the data obtained after subtracting the dead time filtered counts from the unfiltered counts (Fig. 2). While the background counts did not show even a single triplet or higher multiplicity count throughout the 15 day period, there were upto 6 and 7 counts in the

MATERIALS CORNER (CONTINUED FROM PAGE 5)

This issue: TRITIUM PRODUCTION and MORE

100 micro-sec duration in the presence of H_2O and D_2O cells. The burst events were however very few, average value being 1.7, 3.8 and 7.6 bursts per day for the case of background, H_2O cell and D_2O cell respectively.

Protocol for Controlled and Rapid Loading/Unloading of H_2/D_2 Gas from Self-Heated Palladium Wires to Trigger Nuclear Events

A.Garg, R.Rout, M.Srinivasan, T.Sankaranarayanan, A. Shyam and L.Kulkarni, Bhabha Research Centre

Trombay, Bombay, India -

It has now been established that during electrolysis of LiOD using Pd cathodes, a threshold loading ratio of at least 0.85 needs to be achieved before excess heat production can be expected. However, for production of neutrons and tritium (and possibly charged particles and transmutation products too), much lower D/Pd ratios, in the region of 0.4 to 0.7 appear to be adequate. This has independently been corroborated in a variety of gas loading experiments also. It is not so much the magnitude of the steady state loading ration that is attained, but rather the creation of non-equilibrium conditions which facilitates rapid migration/transport of deuterons within the Pd lattice (as demonstrated for example in Yamaguchi's experiments) that seems to be required.

With this in view, a systematic study has been undertaken using electrically self-heated 0.125 mm dia Pd wires in H_2/D_2 atmosphere to optimise the conditions under which rapid loading/deloading of H_2 or D_2 can be achieved. The advantages of self-heated wires are: (a) any desired wire temperature can be obtained by precisely adjusting the current; (b) the instantaneous loading can be deduced from the resistance ratio which can also be independently verified by differential gas pressure measurements. It is found that the absorption rate is rapid if the wire temperature is set just below the

knee of the loading ratio vs temperature curve (approx. 65 deg. C for D_2 and approx. 115 deg. C for H_2), while "instantaneous" desorption is achievable by switching the current (hence wire temperature) to an appropriate high value (> 130 deg. C for D_2 and > 170 deg. C for H_2). With this technique, loading ratios of 0.6 - 0.7, close to the peak region of resistance curve, have been achieved within about 5 - 10 minutes, in a reproducible manner, provided the wire surface is "suitably activated" prior to commencement. Gas pressures used in our studies to date have been < 4 bars.

Having learnt to absorb/desorb D (or H) in a controlled manner in thin Pd wires and create non-equilibrium conditions, efforts are under way to detect various nuclear products such as charged particles (surface barrier detector and CR-39 SSNTDs), neutrons (annular BF_3 counter bank and bubble neutron detectors) tritium (gas flow proportional counter and liquid scintillation counting following dissolution/distillation),

X-rays (photographic films and image intensifier tube with X-ray sensitive phosphor) and radioactivity (Ge-Li detector). The overall objective of the present study is to try and devise a simple experiment which can demonstrate "on demand" the occurrence of at least one anomalous nuclear phenomenon in a deuterated Pd lattice in a reproducible/repeatable manner.

ANSWER: (photo on page 1)

It is an self-photograph of a used electrode obtained by having the generated radioactivity in the electrode expose the photographic film. This autoradiograph is of a palladium hydride wire loaded to ~ 0.6 [after Garg, Srinivasan, et alia.] as described above.

In coming issues:
ENGINEERING DATA
PROTON SEMICONDUCTORS
MORE MATERIAL INFO
HELIUM DIFFUSION
ZERO POINT ENERGIES

"Whatever Happened to 'Cold Fusion'?"

[cont. from p.10]

"They do not support the view that such reactions are ordinary fusion; hence, 'cold fusion' is merely a label and not an assertion of what is happening. It will be argued that government funding of work in this area would not be wasteful, since hydrogen science and technology are accepted fields of inquiry. They form the basis of a large and growing industry in the U.S. and abroad."

IEEE ENDORSES FUSION RESEARCH

The IEEE has endorsed fusion research despite massive shifts described in this issue. These are excerpts of their position from the IEEE-USA Energy Policy Committee Entity Position Statement on Fusion Power.

"The Institute of Electrical and Electronics Engineers - United States Activities (IEEE-USA) endorses research and development in fusion power aimed at a timely demonstration of fusion as a virtually inexhaustible, environmentally attractive and economical power source for base load electrical power generation. We recommend increased funding for R&D in energy technologies, to provide a diverse set of options for efficient end-use and acceptable electricity generation in the near- and long-terms. Our vision emphasizes energy efficiency and conservation, and a diversification of energy sources including solar and renewable energy, advanced nuclear fission technology, and fusion in order to reduce the need for burning fossil fuels.

In its most achievable form, fusion energy comes from the conversion of hydrogen into helium, with the release of energetic helium nuclei and neutrons. A reduction in the total mass of the reacting particles releases energy, in accord with Einstein's theory of relativity.

The potential advantages of fusion power include: (1) universally available and virtually inexhaustible fuel; (2) negligible atmospheric emissions, in contrast to the CO₂ and acid emission from burning fossil fuels; (3) limited impacts on ecological and geophysical processes; and (4) radiological hazards and proliferation risks significantly less than with nuclear fission power.

FOR THE WORLD

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8

(MORE INFORMATION - SEE PAGE 3 TOO)

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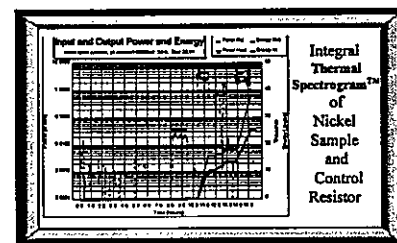
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E-QUEST SCIENCES

(continued from page 6)

Fleischmann gave two lectures. In my opinion, the major important point he made is that heat promotes the CF reaction. This is very important and it has been overlooked by many people in the field even though Fleischmann, Ikegami [24] and other mainstream leaders have pointed it out many times over the years.

Srinivasan reported a number of other extraordinary experiments from various labs at BARC that range from weird to extremely promising. He acts as a representative from India, because not many Indian scientists are able to attend these international conferences. He gave two lectures to cover the work of many other groups. [4, 5] His dynamic and fascinating talks move at such a rapid pace and touch on many amazing topics.

There was an interesting contrast between Kennel, Hagelstein and Smullin [25] on one hand and Karabut [26]. In 1992, Karabut et al. first reported excess heat and gamma rays from a glow discharge experiment. Karabut has improved the heat measurements with a single flow calorimeter, instead of three static calorimeters for each of the three main components. This puts the excess heat on much firmer ground. DuFour, at Shell Research, [19] made the same improvement as Karabut, with equally good results. He combined several separate calorimeters for different component into one unified flow calorimeter, which accounts for all inputs and output. He continues to detect up to 7 watts of excess heat. It is good to see that the oil companies are seriously pursuing this form of energy. Hagelstein has been working hard for the past few years to replicate this experiment, but he has achieved little success. Kennel et al. "call attention to various means by which false positive signals can be observed in x-ray and gamma spectroscopy." They cast doubt upon Karabut's findings: "The authors wonder if other researchers presenting data on gamma and x-ray emission from cold fusion experiments may have fallen prey to similar phenomena. However, it is most expressly not the intention of this paper to make the claim that all positive results are

due to detector artifacts and faulty estimates of statistical significance. . . ."

Another oil company finally came of the woodwork. Amoco reported some old but extremely important early results. Eisner [27], of the University of Houston, described the 1989 experiments that he and Lautzenhiser and Phelps of the Amoco Production Company performed. According to Amoco's 1989 report [28], the first experiment "yielded a 30% energy gain over the life of the experiment (two months). In June 1989, the experiment was modified and a second run also yielded "about 30% excess energy until the catalyst become waterlogged." Other successful runs were performed. Their conclusion: "The calorimetry conclusively shows excess energy was produced within the electrolytic cell over the period of the experiment. This amount, 50 kilojoules, is such that any chemical reaction would have been in near molar amounts to have produced the energy. Chemical analysis shows that no such chemical reactions occurred. The tritium results show that some form of nuclear reactions occurred during the experiment."

As a humorous aside, let me add that when we showed the Amoco results to Morrison at ICCF4 in 1994, after the closing ceremony, he turned pale as a ghost and took off like a shot. This is one of the many nightmare results that "skeptics" wish they could forget, along with KEK, Mitsubishi, NTT, Los Alamos, E-Quest, SRI, Canon, etc., etc.

Hansen [29] described more about his detailed analysis of the 1989 Harwell data, which he previously discussed at ICCF3 and ICCF4. The Harwell experiments were performed in the summer of 1989 by inexperienced junior scientists, who mistakenly concluded that there was

no excess heat. Hansen has more experience with electrochemistry and calorimetry than the Harwell researchers, and he was given full access to their data. The Harwell researchers relied upon hardware which they did not fully understand, and upon a sophisticated analysis which turned out to be unnecessarily complex. Hansen's linear regression "is much simpler and faster" than Harwell's technique. By using better algorithms with the same data, he achieved an order of magnitude better precision than they did. You might call this a triumph of software over hardware. The Harwell experiments are not intrinsically important. Much better results have been achieved since 1989, and far better techniques have been discovered, so there is no technological imperative to look back at these early results. But for political reasons it is important to show that the conclusions published by Harwell were incorrect.

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MORE REPORTS:

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Much has been going on.

LECTURE: "Whatever Happened to 'Cold Fusion'?"

Washington, D.C. - David Nagel, Superintendent of the Condensed Matter and Radiation Sciences Division, U. S. Naval Research Laboratory (NRL), scheduled a lecture on cold fusion at The Philosophical Society of Washington 2048th Meeting Friday, October 20, 1995. The Philosophical Society, founded 1871, took a quantum leap in its knowledge of this field in its second talk on cold fusion; the Society previously hosted Talbott Chubb. The following are excerpts from the abstract of the lecture:

"Cold Fusion' is now ignored, disdained and even mocked by scientists and the public. This is due to a variety of mistakes by scientists and the government, and the unwillingness of journal, magazine and newspaper editors to pay attention to this topic. But, does this mean that nothing is being done, or should be done, in response to questions raised by work on 'cold fusion'? Several hundred peopleworld-wide are spending full or part time trying to get to the bottom of the mysteries which followed from the 1989 announcement by Pons and Fleischmann. The fifth international conference in Europe earlier this year attracted 200 people, half of them with industrial connections, mainly from Japan, Italy, France and the U.S.

There are active programs in Russia, India and China. A framework for organizing work on 'cold fusion' will be presented, along with some of the data which cannot, in the opinion of the speaker, be attributed to fraud or error. If correct, these data would strongly indicate that nuclear reactions are behind some of the observations. "

(Continued on page 8)

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